

SEMESTER – III

COURSE: SKILL ENHANCEMENT COURSE(SEC-III)

PAPER: COMPUTER SOFTWARE, PROGRAMMING & AI

TOTAL CREDITS: THEORY-03

TEACHING HOURS: THEORY-45

EVALUATION (Only the End Semester University Examination will be conducted)	
Full Marks	75 Marks
Duration of Exam	3 Hours
Pass Marks	30 Marks

COURSE OBJECTIVES: This course objective is to provide a solid foundation in the concepts and components of computer software systems. To equip students with practical programming knowledge using Python. To introduce core concepts of Artificial Intelligence and its integration with real-world. To foster awareness of the societal impacts of emerging AI technologies.

COURSE OUTCOMES: Identify and differentiate between types of software. Python program's using fundamental and intermediate programming constructs. Understand the theoretical foundation of AI. Analyze the influence of AI technologies on industries, society, and decision-making.

COURSE CONTENTS:

UNIT-I: Introduction to Computer Software

Introduction to Computer Software: Importance, Classification, Application Software vs Utility Software, Software Development Life Cycle (SDLC). Phases: Requirement Analysis, Design, Implementation, Testing, Deployment, Maintenance Translators: Compilers, Interpreters, Assembler

UNIT-II: Introduction to Python Programming and Object-Oriented Concepts

Introduction to Python: Features, Installation, Syntax, First Python Program, Variables, Keywords, and Data Types, Input/Output. operations and type conversion. Operators: Arithmetic, Logical, Relational, Bitwise. Conditional Statements: if, if-else, elif. Looping Constructs: for, while, break, continue.

Introduction to Object-Oriented Programming: Classes and Objects, Constructors, self keyword, Inheritance, Polymorphism, Encapsulation Searching and Sorting Algorithms: Linear Search, Binary Search, Bubble Sort, Insertion Sort, Selection Sort (with Python implementation).

UNIT-III: Introduction to AI

Introduction to AI: History & Evolution of AI, Definition, Scope, Goals & Types of AI: Narrow, General, Super AI,

Subfields of AI, Applications of AI: Healthcare, Finance, Education, Transportation, Agriculture. Ethical consideration in AI, Intelligent Agent, Agent architecture, Environment type, Rational Agent.

SUGGESTED READINGS:

1. “Learning Python” by Mark Lutz (O’ Reilly)
2. “Artificial Intelligence: A Guide for Thinking Humans” by Melanie Mitchell.
3. “Code: The Hidden language of Computer Hardware and Software” by Charles Petzold.
4. “Artificial Intelligenc” By Saroj Kaushik, publisher- Cengage.

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